

DTS103TC

Data Structure and Algorithms

Lab 1: Getting Started

Dr. Qi Chen, Dr. Di Zhang, Dr. Md Maruf Hasan,
Dr. Guangyu Ren and Dr. Bin Chen
School of AI and Advanced Computing

Learning outcomes

- Python + Jupyter Notebook
- Insertion Sort
- Selection Sort
- Asymptotic analysis of algorithms
- Lab Exercises

Python and Jupyter Notebook

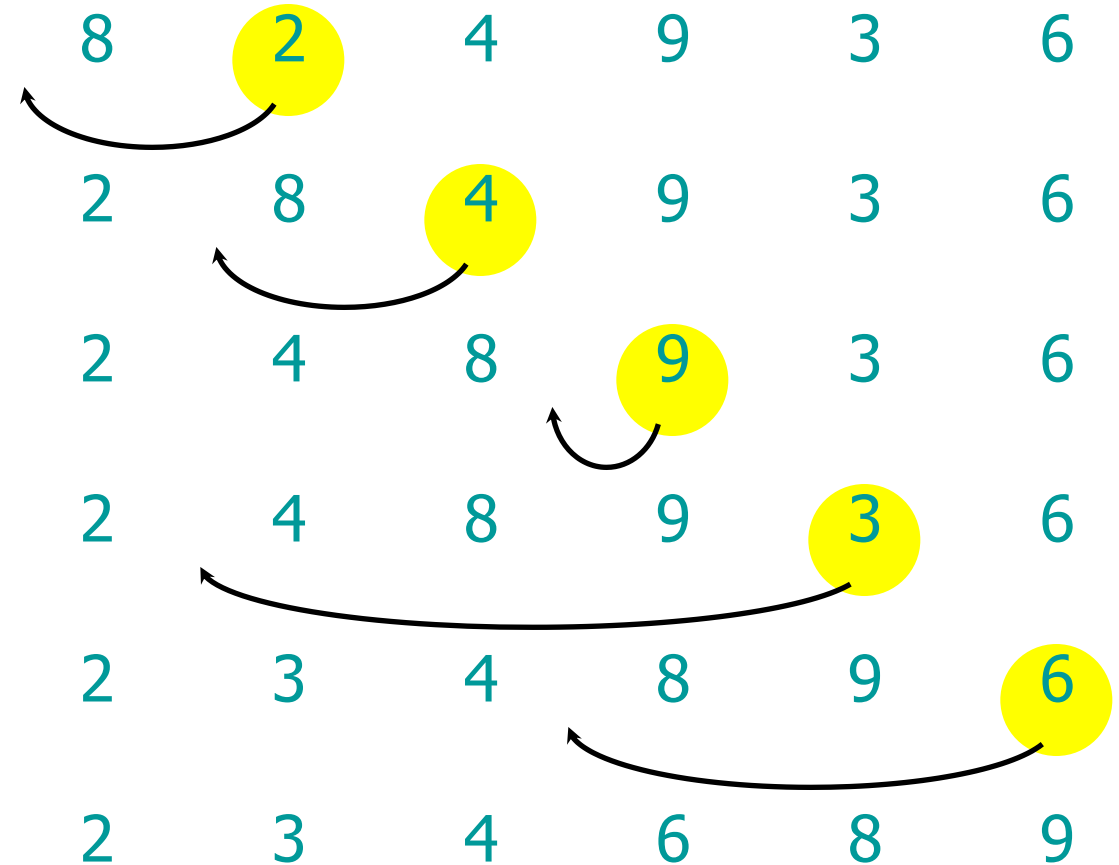
- We assume you can do Python programming
- If you are not familiar with Python, or need some refresher exercises. You can try the Kaggle python course:
<https://www.kaggle.com/learn/python>
- Jupyter notebook is very convenient for demonstrating and managing your coursework and projects
- <https://jupyter.org/install>

Insertion Sort

Insertion sort

INSERTION-SORT(A)

```
1  for  $j = 2$  to  $A.length$ 
2       $key = A[j]$ 
3      // Insert  $A[j]$  into the sorted sequence  $A[1..j-1]$ .
4       $i = j - 1$ 
5      while  $i > 0$  and  $A[i] > key$ 
6           $A[i + 1] = A[i]$ 
7           $i = i - 1$ 
8       $A[i + 1] = key$ 
```



Insertion sort

INSERTION-SORT(A)

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4       $i = j - 1$ 
5      while  $i > 0$  and  $A[i] > key$ 
6           $A[i+1] = A[i]$ 
7           $i = i - 1$ 
8       $A[i+1] = key$ 
```

```
def insertion_sort(InputList):
    for j in range(1, len(InputList)):
        key = InputList[j]
        i = j-1
        # Compare the current element with next one
        while (InputList[i] > key) and (i >= 0):
            InputList[i+1] = InputList[i]
            i=i-1
        InputList[i+1] = key

arr = [19,2,31,45,30,11,121,27]

insertion_sort(arr)

print(arr)
```

[2, 11, 19, 27, 30, 31, 45, 121]

Selection Sort

Selection sort

- sort (34, 10, 64, 51, 32, 21) in ascending order

Sorted part	Unsorted part	Swapped
	34 10 64 51 32 21	10, 34
10	34 64 51 32 21	21, 34
10 21	64 51 32 34	32, 64
10 21 32	51 64 34	51, 34
10 21 32 34	64 51	51, 64
10 21 32 34 51	64	--
10 21 32 34 51 64		

Selection sort

```
for i = 1 to n-1:  
    min = i  
    for j = i+1 to n do  
        if a[j] < a[min]  
            min = j  
    swap a[i] and a[min]
```

Selection sort

- Worst-case Time complexity?
- Best-case time complexity?
- Average-case time complexity?
- (Auxiliary) space complexity?

Selection sort

- Worst-case Time complexity: $O(n^2)$
- Best-case time complexity: $O(n^2)$
- Average-case time complexity: $O(n^2)$
- (Auxiliary) space complexity: $O(1)$

Time complexity of this?

```
for (i=0;i<n;i++)  
{  
    stmt  
}
```

O(?)

O(n)

Time complexity of this?

```
for (i=n;i>0;i--)  
{  
    stmt  
}
```

O(?)

O(n)

Time complexity of this?

```
for (i=0;i<n;i=i+2)
{
    stmt
}
```

O(?)

O(n)

Time complexity of this?

```
for (i=0;i<n;i++)  
    for (j=0;j<n;j++)  
    {  
        stmt  
    }
```

$O(?)$

$O(n^2)$

Time complexity of this?

```
for (i=0;i<n;i++)  
{  
    stmt  
}  
for (j=0;j<n;j++)  
{  
    stmt  
}
```

$O(?)$

$O(n)$

Time complexity of this?

```
for (i=0;i<n;i++)  
    for (j=0;j<i;j++)  
    {  
        stmt  
    }
```

$O(?)$

$O(n^2)$

Time complexity of this?

```
j=0
for (i=0;j<n;i++)
{
    j=j+i
}
```

$O(?)$

$O(\sqrt{n})$

Time complexity of this?

```
for (i=1;i<n;i=i*2)
{
    stmt
}
```

$O(?)$

$O(\log n)$

Time complexity of this?

```
k=0
for (i=1;i<n;i=i*2)
{
    k++;
}
for (j=1;j<k;j=j*2)
{
    stmt
}
```

$O(?)$

$O(\log n + \log \log n)$

$= O(\log n)$

Time complexity of this?

```
for (i=0;i<n;i++)  
{  
    for (j=1;j<n;j=j*2)  
    {  
        stmt  
    }  
}
```

O(?)

O(nlogn)

Some algorithms we learnt

INSERTION-SORT(A)

```
1  for  $j = 2$  to  $A.length$ 
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4       $i = j - 1$ 
5      while  $i > 0$  and  $A[i] > key$ 
6           $A[i+1] = A[i]$ 
7           $i = i - 1$ 
8       $A[i+1] = key$ 
```

$O(?)$

$O(n^2)$

Some algorithms we learnt

```
for i = 1 to n-1:  
    min = i  
    for j = i+1 to n do  
        if a[j] < a[min]  
            min = j  
    swap a[i] and a[min]
```

$O(?)$

$O(n^2)$

Exercises for Lab 1

- Environment Setup
- Implement Insertion Sort
- Implement Selection Sort
- Implement Bubble Sort
- Generate arrays to test your functions